

- Part 1 was completely done in python. the code and image is the following

:

```

- import matplotlib.pyplot as plt
  v1v2list = [
    (0, 0),
    (3.02, 0.71),
    (6.03, 0.73),
    (9.06, 0.74),
    (12.00, 0.75),
    (15.05, 0.76),
    (18.03, 0.76),
    (20.9, 0.77),
    (24.0, 0.77),
    (26.9, 0.77),
    (29.9, 0.78),
    (-3.0, -2.99),
    (-6.0, -4.23),
    (-9.0, -4.54),
    (-12.0, -4.68),
    (-15.0, -4.77),
    (-18.0, -4.83),
    (-21.0, -4.87),
    (-24.0, -4.90),
    (-27.0, -4.93),
    (-30.0, -4.95),
    (1.5, 0.68),
    (1.0, 0.66),
  ]
  v1v2list.sort(key=lambda x: x[0])
  v1list = [x[0] for x in v1v2list]
  v2list = [x[1] for x in v1v2list]
  currentlist = [(x[0] - x[1]) / 3900 for x in v1v2list]

  v1listuncertainty = [ufloat(x[0], abs(x[0] * 0.012)) for x in v1v2list]
  v2listuncertainty = [ufloat(x[1], abs(x[1] * 0.012)) for x in v1v2list]
  currentlistuncertainty = [
    ((x[0] - x[1]) / 3900).s
    for x in [
      (v1listuncertainty[i], v2listuncertainty[i]) for i in range(len(v1v2list))
    ]
  ]

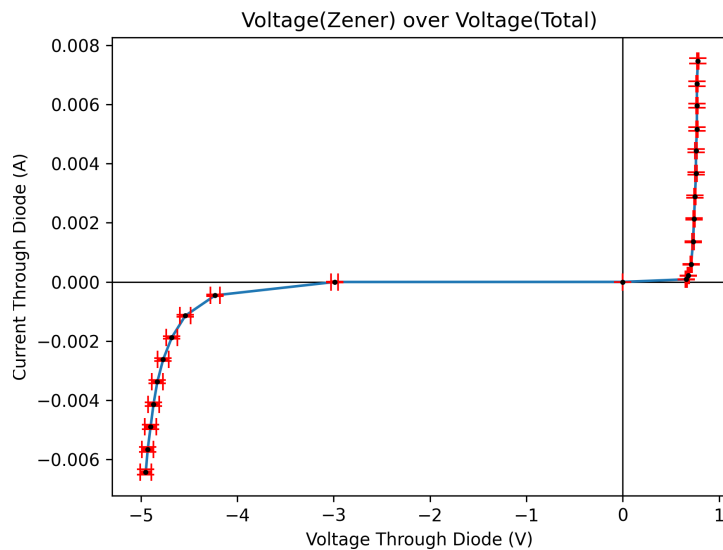
  plt.axhline(0, color="black", linewidth=0.8)
  plt.axvline(0, color="black", linewidth=0.8)
  plt.plot(v2list, currentlist)
  plt.errorbar(
    v2list,

```

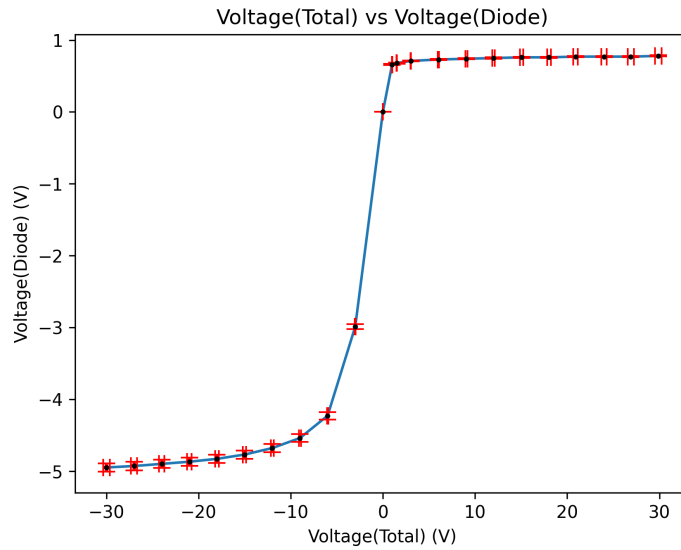
```

currentlist,
xerr=[abs(x) * 0.012 for x in v2list],
yerr=[y for y in currentlistuncertainty],
fmt="o",
color="black",
ecolor="red",
capsize=5,
elinewidth=2,
markersize=2,
)
plt.title("Voltage(Zener) over Voltage(Total)")
plt.xlabel("Voltage Through Diode (V)")
plt.ylabel("Current Through Diode (A)")
plt.savefig("currentvoltage", dpi=300)
plt.clf()

```



-
- Estimating the knee, I'd say that the [[forward]] bias voltage tapers off at around $0.6\text{-}0.7 \pm 0.1$, for brevity lets say 0.65 ± 0.1
- On the [[other]] side, I'd also estimate the backwards bias voltage tapers off at around -5 ± 0.1
- There is also another visual of total voltage vs voltage through the diode.



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- we also see tapering at around the 0.65 and -5.

- Part 2

—

$$I_{Total} = \frac{V_{Total}}{R_{Total}}$$

—

$$I_{Total} = \frac{V_{Total}}{(1k\Omega + 1k\Omega)}$$

—

$$V_{R_L} = I_{Total} R_{R_L}$$

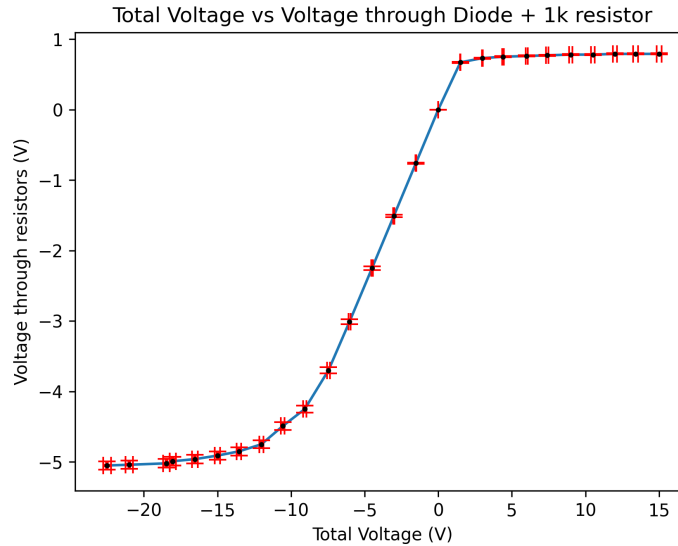
—

$$V_{R_L} = \frac{V_{Total}}{(1k\Omega + 1k\Omega)} \times (1k\Omega)$$

—

$$V_{R_L} = \frac{V_{Total}}{2}$$

- $\therefore R_L$ is one half of the supply voltage



- You can see the graph of the voltage over the diode tapers off. for example for the [[forward]] bias, the diode only has a voltage drop of $\sim 0.6 \pm 0.1V$, before it tapers off, similarly for the negative bias, it tapers off at around $\sim -5 \pm 0.1V$
- One thing to note is we measured in sizes of $\sim 1.5v$, and we decided to go to -24 volts because our graph didn't properly show the extent of the tapering off.
- And the for the above graph ('Total Voltage vs Voltage through Diode + 1k resistor'):

```
* part2vlist = [ (0, 0), (1.5, 0.67), (3, 0.73), (4.4, 0.75), (6, 0.76),
(7.4, 0.77), (9, 0.78), (10.5, 0.78), (12.0, 0.79), (13.4, 0.79), (15,
0.79), (-1.52, -0.76), (-3.02, -1.51), (-4.50, -2.25), (-6.03, -3.01),
(-7.46, -3.70), (-9.07, -4.25), (-10.56, -4.49), (-12.02, -4.75), (-13.54,
-4.85), (-15, -4.91), (-16.54, -4.96), (-18.04, -4.99), (-18.04, -4.99),
(-18.46, -5.02), (-21.0, -5.04), (-22.5, -5.05), ]
part2vlist.sort(key=lambda x: x[0])
```

```
v1list = [x[0] for x in part2vlist]
v2list = [x[1] for x in part2vlist]
```

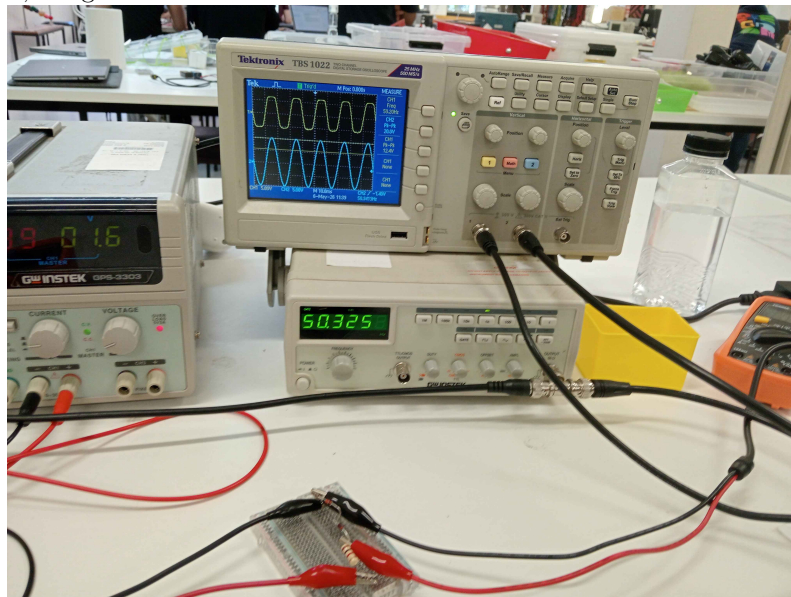
```
plt.plot(v1list, v2list)
plt.errorbar(
    v1list,
    v2list,
    xerr=[abs(x) * 0.012 for x in v1list],
    yerr=[abs(y) * 0.012 for y in v2list],
    fmt="o",
```

```

        color="black",
        ecol="red",
        capsize=5,
        elinewidth=2,
        markersize=2,
    )
    plt.title("Total Voltage vs Voltage through Diode + 1k resistor")
    plt.xlabel("Total Voltage (V)")
    plt.ylabel("Voltage through resistors (V)")
    plt.savefig("part2", dpi=300)

```

- Part 3, image below.



- Using the ‘measure’ button on the oscilloscope, we changed the amplitude until it was ~20V raw, and the through the oscilloscope measurement, using ‘pk-pk’, we obtain a number of $12.4V \pm 0.1$. Pk-Pk doesn’t really have an uncertainty value, so arbitrarily i chose 0.1 as the uncertainty.
- This is the voltage drop over the two zener diodes, the amplitude of the wave is equal to the [[forward]] and negative bias of the oscilloscope. The amplitude is also 1/2 the pk-pk value. so:
 - * $\frac{PkPk}{2} = Amplitude = \frac{12.4 \pm 0.1}{2}$
 - * from uncertainties import ufloat
 - amplitude = ufloat(12, 0.1)
 - print(amplitude / 2)
 - * Output is $6 \pm 0.05V$
- using part two, we will use:
 - * $Amplitude = V_{ForwardBias} + V_{NegativeBias}$
 - * which is $(5 \pm 0.1) + (0.6 \pm 0.1)$

- ```
* vforwardbias = ufloat(5, 0.1)
 vnegativebias = ufloat(0.6, 0.1)
 print(vforwardbias + vnegativebias)
```
- \* the output is  $5.6 \pm 0.14$
- Comparing our part `[[2]]` values and our measured value for the maximum value, its clear there is a discrepancy, as in the uncertainty ranges of both of these values don't match.
  - But our values are similar, 5.6 and 6 are close values.
  - I don't have a theory as to why we have a systematic error here but it does exist.